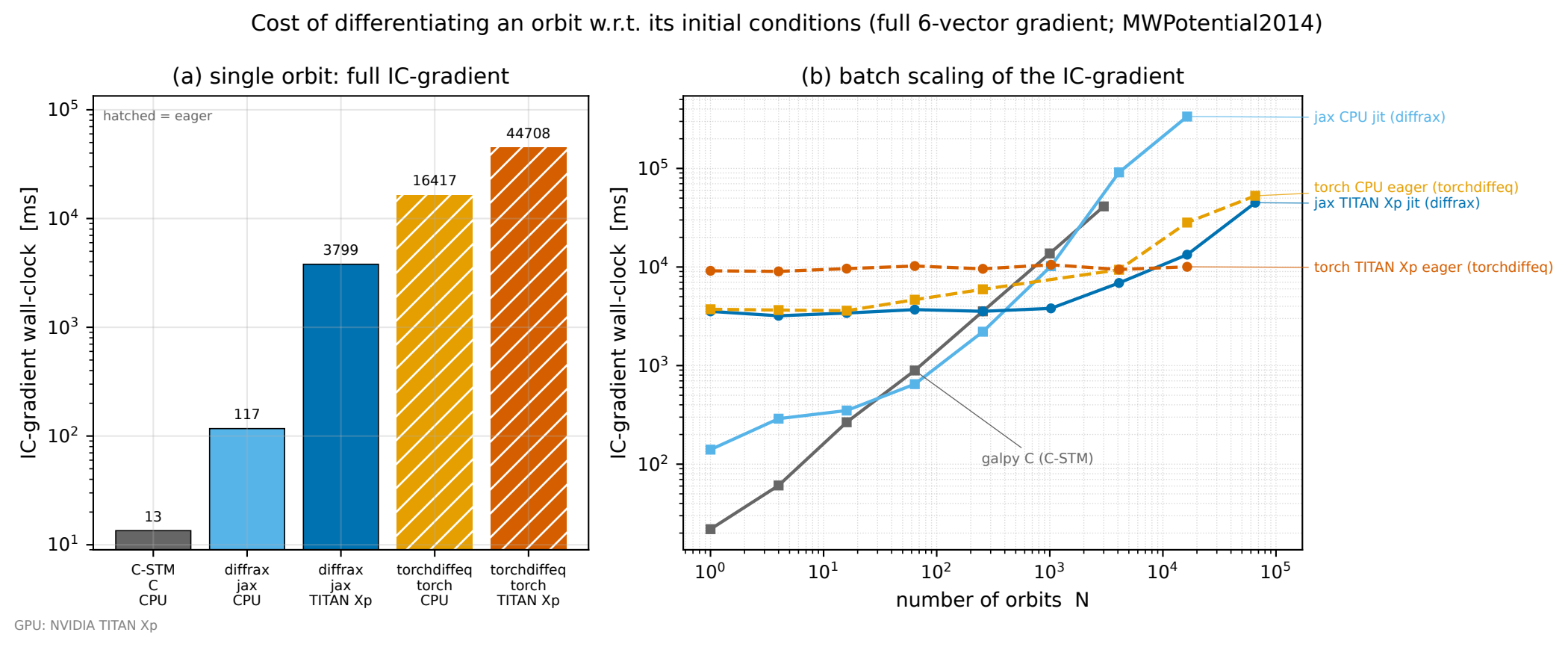
Cost of differentiating an orbit w.r.t. its initial conditions (full 6-vector gradient; MWPotential2014)
(a) single orbit: full IC-gradient
hatched = eager
IC-gradient wall-clock [ms]
13
117
3799
16417
44708
C-STM
C
CPU
diffrax
jax
CPU
diffrax
jax
TITAN Xp
torchdiffeq
torch
CPU
torchdiffeq
torch
TITAN Xp
GPU: NVIDIA TITAN Xp
(b) batch scaling of the IC-gradient
IC-gradient wall-clock [ms]
number of orbits N
jax CPU jit (diffrax)
torch CPU eager (torchdiffeq)
jax TITAN Xp jit (diffrax)
torch TITAN Xp eager (torchdiffeq)
galpy C (C-STM)